

Lab 35: Pandas Filtering and Sorting

Aim

To filter DataFrame rows based on conditions and sort the results using one or more columns.

Algorithm

1. Import the Pandas library.
2. Create a DataFrame containing student details.
3. Apply multiple conditions using boolean indexing.
4. Filter the rows that satisfy the conditions.
5. Sort the DataFrame using multiple columns.
6. Display the filtered and sorted results.

Program

Lab 35: Pandas Filtering and Sorting

```
import pandas as pd
```

```
# Create DataFrame
```

```
df = pd.DataFrame({  
    "name": ["A", "B", "C", "D"],  
    "class": ["I", "II", "I", "II"],  
    "marks": [78, 85, 92, 66]  
})
```

```
# Filter using multiple conditions
```

```
filtered = df[  
    (df["marks"] > 70) &  
    (df["class"] == "I")  
]
```

```
print("Filtered:\n", filtered)
```

```
# Sort by class and marks
```

```
sorted_df = df.sort_values(  
    ["class", "marks"],  
    ascending=[True, False]  
)
```

```
print("\nSorted:\n", sorted_df)
```

Sample Output

Filtered:

	name	class	marks
0	A	I	78
2	C	I	92

Sorted:

	name	class	marks
2	C	I	92
0	A	I	78
1	B	II	85
3	D	II	66

Result

Thus, multi-condition filtering and multi-column sorting were successfully performed on a Pandas DataFrame.